Hollow Sections Sectional Properties

H	B	T	M	A	I_{xx}	L_{yy}	i_{xx}	i_{yy}	Wel_{xx}	Wel_{yy}	Wpl_{xx}	Wpl_{yy}	Lt	Ct	As		d	d
Size		Thickness	Mass per unit length	Cross sectional area	Second moment of area	Second moment of area	Radius of gyration	Radius of gyration	Elastic section modulus	Elastic section modulus	Plastic section modulus	Plastic section modulus	Torsional inertia constant	Torsional modulus constant	Superarea per metre length	Nominal length per tonne		
H	B	T	M	A	I_{xx}	L_{yy}	i_{xx}	i_{yy}	Wel_{xx}	Wel_{yy}	Wpl_{xx}	Wpl_{yy}	Lt	Ct	As		d	d
mm	mm	mm	kg/m	cm^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3	cm^4	cm^3	m^2/m	m	cm	cm
40	20	2	1.68	2.14	4.05	1.34	1.38	0.793	2.02	1.34	2.61	1.6	3.45	2.36	0.113	596	36	16
40	20	2.5	2.03	2.59	4.69	1.54	1.35	0.77	2.35	1.54	3.09	1.88	4.06	2.72	0.111	492	35	15
40	20	3	2.36	3.01	5.1	1.68	1.32	0.748	2.6	1.68	3.5	2.12	4.57	3	0.11	423	34	14
50	30	2	2.31	2.94	9.51	4.29	1.8	1.21	3.81	2.86	4.74	3.33	9.77	4.84	0.153	434	46	26
50	30	2.5	2.82	3.59	11.3	5.05	1.77	1.19	4.52	3.37	5.7	3.98	11.7	5.72	0.151	355	45	25
50	30	3	3.3	4.21	12.8	5.7	1.75	1.16	5.13	3.8	6.57	4.58	13.5	6.49	0.15	303	44	24
50	30	4	4.2	5.35	15.3	6.69	1.69	1.12	6.1	4.46	8.05	5.58	16.5	7.71	0.146	238	42	22
60	40	2	2.93	3.74	18.4	9.83	2.22	1.62	6.14	4.92	7.47	5.65	20.7	8.12	0.193	341	56	36
60	40	2.5	3.6	4.59	22.1	11.7	2.19	1.6	7.36	5.87	9.06	6.84	25.1	9.72	0.191	278	55	35
60	40	3	4.25	5.41	25.4	13.4	2.17	1.58	8.46	6.72	10.5	7.94	29.3	11.2	0.19	236	54	34
60	40	4	5.45	6.95	31	16.3	2.11	1.53	10.3	8.14	13.2	9.89	36.7	13.7	0.186	183	52	32
60	40	5	6.56	8.36	35.3	18.4	2.06	1.48	11.8	9.21	15.4	11.5	42.8	15.6	0.183	152	50	30
70	50	2	3.56	4.54	31.5	18.8	2.63	2.03	8.99	7.5	10.8	8.58	37.5	12.2	0.233	281	66	46
70	50	2.5	4.39	5.59	38	22.6	2.61	2.01	10.9	9.04	13.2	10.4	45.8	14.7	0.231	228	65	45
70	50	3	5.19	6.61	44.1	26.1	2.58	1.99	12.6	10.4	15.4	12.2	53.6	17.1	0.23	193	64	44
70	50	4	6.71	8.55	54.7	32.2	2.53	1.94	15.6	12.9	19.5	15.4	68.1	21.2	0.226	149	62	42
70	50	5	8.13	10.4	63.5	37.2	2.48	1.9	18.1	14.9	23.1	18.2	80.8	24.6	0.223	123	60	40
80	40	2	3.56	4.54	37.4	12.7	2.87	1.67	9.34	6.36	11.6	7.17	30.9	11	0.233	281	76	36
80	40	2.5	4.39	5.59	45.1	15.3	2.84	1.65	11.3	7.63	14.1	8.72	37.6	13.2	0.231	228	75	35
80	40	3	5.19	6.61	52.3	17.6	2.81	1.63	13.1	8.78	16.5	10.2	43.9	15.3	0.23	193	74	34
80	40	4	6.71	8.55	64.8	21.5	2.75	1.59	16.2	10.7	20.9	12.8	55.2	18.8	0.226	149	72	32
80	40	5	8.13	10.4	75.1	24.6	2.69	1.54	18.8	12.3	24.7	15	65	21.7	0.223	123	70	30
80	60	2	4.19	5.34	49.5	31.9	3.05	2.44	12.4	10.6	14.7	12.1	61.2	17.1	0.273	239	76	56
80	60	2.5	5.17	6.59	60.1	38.6	3.02	2.42	15	12.9	18	14.8	75.1	20.7	0.271	193	75	55
80	60	3	6.12	7.81	70	44.9	3	2.4	17.5	15	21.2	17.4	88.3	24.1	0.27	163	74	54
80	60	4	7.97	10.1	87.9	56.1	2.94	2.35	22	18.7	27	22.1	113	30.3	0.266	126	72	52
80	60	5	9.7	12.4	103	65.7	2.89	2.31	25.8	21.9	33.2	26.4	136	35.7	0.263	103	70	50
90	50	2	4.19	5.34	57.9	23.4	3.29	2.09	12.9	9.35	15.7	10.5	53.4	15.9	0.273	239	86	46
90	50	2.5	5.17	6.59	70.3	28.2	3.27	2.07	15.6	11.3	19.3	12.8	65.3	19.2	0.271	193	85	45
90	50	3	6.13	7.81	81.9	32.7	3.24	2.05	18.2	13.1	22.6	15	76.7	22.4	0.27	163	84	44
90	50	4	7.97	10.1	103	41	3.18	2	22.8	16.3	28.8	19.1	97.7	28	0.266	126	82	42
90	50	5	9.7	12.4	121	47	3.12	1.96	26.8	18.9	34.4	22.7	116	32.7	0.263	103	80	40
100	40	2.5	5.17	6.59	79	19	3.47	1.69	15.9	9.39	20.2	10.6	50.5	16.8	0.271	193	95	35
100	40	3	6.13	7.81	92.3	21.7	3.44	1.67	18.5	10.8	23.7	12.4	59	19.4	0.27	163	94	34
100	40	4	7.97	10.1	116	26.7	3.38	1.62	23.1	13.3	30.3	15.7	74.5	24	0.266	126	92	32
100	40	5	9.7	12.4	136	30.8	3.31	1.58	27.1	15.4	36.1	18.5	87.9	27.9	0.263	103	90	30
100	50	2.5	5.56	7.09	91.2	31.1	3.59	2.09	18.2	12.4	22.7	14	75.4	21.5	0.291	180	95	45

100	50	3	6.6	8.41	106	36.1	3.56	2.07	21.3	14.4	26.7	16.4	88.6	25	0.29	152	94	44
100	50	4	8.59	10.9	134	44.9	3.5	2.03	26.8	18	34.1	20.9	113	31.3	0.286	116	92	42
100	50	5	10.5	13.4	158	52.5	3.44	1.98	31.6	21	40.8	25	135	36.8	0.283	95.4	90	40
100	50	6	12.3	15.6	179	58.7	3.38	1.94	35.8	23.5	46.9	28.5	154	41.4	0.279	81.5	88	38
100	50	6.3	12.5	15.9	176	58.2	3.32	1.91	35.1	23.3	46.9	28.6	158	42.1	0.273	79.9	87.4	37.4
100	60	2.5	5.96	7.6	103	46.9	3.69	2.49	20.6	15.6	25.1	17.7	103	26.2	0.311	168	95	55
100	60	3	7.07	9.01	121	54.6	3.66	2.46	24.1	18.2	29.6	20.8	122	30.6	0.31	141	94	54
100	60	4	9.22	11.7	153	68.7	3.6	2.42	30.5	22.9	37.9	26.6	156	38.7	0.306	108	92	52
100	60	5	11.3	14.4	181	80.8	3.55	2.37	36.2	26.9	45.6	31.9	188	45.8	0.303	88.7	90	50
100	60	6	13.2	16.8	205	91.2	3.49	2.33	41.1	30.4	52.5	36.6	216	51.9	0.299	75.7	88	48
100	60	6.3	13.5	17.2	203	90.9	3.44	2.3	40.7	30.3	52.8	36.9	223	53	0.293	74	87.4	47.4
100	80	2.5	6.74	8.59	127	90.2	3.84	3.24	25.4	22.5	30	25.8	166	35.7	0.351	148	95	75
100	80	3	8.01	10.2	149	106	3.82	3.22	29.8	26.4	35.4	30.4	196	41.9	0.35	125	94	74
100	80	4	10.5	13.3	189	134	3.77	3.17	37.9	33.5	45.6	39.2	254	53.4	0.346	95.4	92	72
100	80	5	12.8	16.4	226	160	3.72	3.12	45.2	39.9	55.1	47.2	308	63.7	0.343	77.9	90	70
100	80	6	15.1	19.2	258	182	3.67	3.08	51.7	45.5	63.8	54.7	357	73	0.339	66.2	88	68
100	80	6.3	15.5	19.7	259	183	3.62	3.04	51.8	45.7	64.4	55.4	371	75	0.333	64.6	87.4	67.4
120	60	2.5	6.74	8.59	161	55.2	4.33	2.53	26.9	18.4	33.2	20.6	133	31.7	0.351	148	115	55
120	60	3	8.01	10.2	189	64.4	4.3	2.51	31.5	21.5	39.2	24.2	156	37.1	0.35	125	114	54
120	60	4	10.5	13.3	241	81.2	4.25	2.47	40.1	27.1	50.5	31.1	201	47	0.346	95.4	112	52
120	60	5	12.8	16.4	287	96	4.19	2.42	47.8	32	60.9	37.4	242	55.8	0.343	77.9	110	50
120	60	6	15.1	19.2	328	109	4.13	2.38	54.7	36.3	70.6	43.1	280	63.6	0.339	66.2	108	48
120	60	6.3	15.5	19.7	327	109	4.07	2.35	54.5	36.4	71.2	43.7	289	65.1	0.333	64.6	107.4	47.4
120	60	8	18.9	24	375	124	3.95	2.27	62.6	41.3	84.1	51.3	340	75	0.326	53	104	44
120	80	3	9	11.4	230	123	4.49	3.29	38.4	30.9	46.2	35	255	50.8	0.39	112	114	74
120	80	4	11.7	14.9	295	157	4.44	3.24	49.1	39.3	59.8	45.2	331	64.9	0.386	85.2	112	72
120	80	5	14.4	18.4	353	188	4.39	3.2	58.9	46.9	72.4	54.7	402	77.8	0.383	69.4	110	70
120	80	6	17	21.6	406	215	4.33	3.15	67.7	53.8	84.3	63.5	469	89.4	0.379	58.9	108	68
120	80	6.3	17.5	22.2	408	217	4.28	3.12	68.1	54.3	85.6	64.7	488	92.1	0.373	57.3	107.4	67.4
120	80	8	21.4	27.2	476	252	4.18	3.04	79.3	62.9	102	76.9	584	108	0.366	46.8	104	64
140	80	4	13	16.5	430	180	5.1	3.3	61.4	45.1	75.5	51.3	412	76.2	0.426	77	132	72
140	80	5	16	20.4	517	216	5.04	3.26	73.9	54	91.8	62.2	501	91.8	0.423	62.6	130	70
140	80	6	18.9	24	597	248	4.98	3.21	85.3	62	107	72.4	584	106	0.419	53	128	68
140	80	6.3	19.4	24.8	603	251	4.93	3.19	86.1	62.9	109	74	609	109	0.413	51.4	127.4	67.4
140	80	8	23.9	30.4	708	293	4.82	3.1	101	73.3	131	88.4	731	129	0.406	41.8	124	64
150	100	4	14.9	18.9	595	319	5.6	4.1	79.3	63.7	95.7	72.5	662	105	0.486	67.2	142	92
150	100	5	18.3	23.4	719	384	5.55	4.05	95.9	76.8	117	88.3	809	127	0.483	54.5	140	90
150	100	6	21.7	27.6	835	444	5.5	4.01	111	88.8	137	103	948	147	0.479	46.1	138	88
150	100	6.3	22.4	28.5	848	453	5.45	3.98	113	90.5	140	106	992	152	0.473	44.6	137.4	87.4
150	100	8	27.7	35.2	1008	536	5.35	3.9	134	107	169	128	1206	182	0.466	36.1	134	84
150	100	10	33.4	42.6	1162	614	5.22	3.8	155	123	199	150	1426	211	0.457	29.9	130	80
150	100	12	37.7	48.1	1207	642	5.01	3.65	161	128	215	163	1573	229	0.438	26.5	126	76
150	100	12.5	38.9	49.5	1225	651	4.97	3.63	163	130	220	166	1606	233	0.436	25.7	125	75
160	80	4	14.2	18.1	598	204	5.74	3.35	74.7	50.9	92.9	57.4	494	88	0.466	70.2	152	72

160	80	5	17.5	22.4	722	244	5.68	3.3	90.2	61	113	69.7	601	106	0.463	57	150	70
160	80	6	20.7	26.4	836	281	5.62	3.26	105	70.2	132	81.3	702	122	0.459	48.2	148	68
160	80	6.3	21.4	27.3	846	286	5.57	3.24	106	71.4	135	83.3	732	126	0.453	46.7	147.4	67.4
160	80	8	26.4	33.6	1001	335	5.46	3.16	125	83.7	163	100	882	150	0.446	37.9	144	64
160	80	10	31.8	40.6	1146	380	5.32	3.06	143	95	191	117	1031	172	0.437	31.4	140	60
160	80	12	35.8	45.7	1171	391	5.06	2.93	146	97.8	204	125	1111	183	0.418	27.9	136	56
160	80	12.5	36.9	47	1185	396	5.02	2.9	148	98.9	208	127	1129	185	0.416	27.1	135	55
180	100	4	16.8	21.3	926	374	6.59	4.18	103	74.8	126	84	854	127	0.546	59.7	172	92
180	100	5	20.7	26.4	1124	452	6.53	4.14	125	90.4	154	103	1045	154	0.543	48.3	170	90
180	100	6	24.5	31.2	1310	524	6.48	4.1	146	105	181	120	1227	179	0.539	40.8	168	88
180	100	6.3	25.4	32.3	1335	536	6.43	4.07	148	107	186	124	1283	185	0.533	39.4	167.4	87.4
180	100	8	31.4	40	1598	637	6.32	3.99	178	127	226	150	1565	222	0.526	31.8	164	84
180	100	10	38.1	48.6	1859	736	6.19	3.89	207	147	268	177	1859	260	0.517	26.2	160	80
180	100	12	43.4	55.3	1965	782	5.96	3.76	218	156	292	194	2073	285	0.498	23.1	156	76
180	100	12.5	44.8	57	2001	796	5.92	3.74	222	159	300	199	2122	290	0.496	22.3	155	75
200	100	4	18	22.9	1200	411	7.23	4.23	120	82.2	148	91.7	985	142	0.586	55.5	192	92
200	100	5	22.3	28.4	1459	497	7.17	4.19	146	99.4	181	112	1206	172	0.583	44.9	190	90
200	100	6	26.4	33.6	1703	577	7.12	4.14	170	115	213	132	1417	200	0.579	37.9	188	88
200	100	6.3	27.4	34.8	1739	591	7.06	4.12	174	118	219	135	1483	208	0.573	36.6	187.4	87.4
200	100	8	33.9	43.2	2091	705	6.95	4.04	209	141	267	165	1811	250	0.566	29.5	184	84
200	100	10	41.3	52.6	2444	818	6.82	3.94	244	164	318	195	2154	292	0.557	24.2	180	80
200	100	12	47.1	60.1	2607	876	6.59	3.82	261	175	350	215	2414	322	0.538	21.2	176	76
200	100	12.5	48.7	62	2659	892	6.55	3.79	266	178	359	221	2474	329	0.536	20.5	175	75
200	120	4	19.3	24.5	1353	618	7.43	5.02	135	103	164	115	1345	172	0.626	51.9	192	112
200	120	5	23.8	30.4	1649	750	7.37	4.97	165	125	201	141	1652	210	0.623	42	190	110
200	120	6	28.3	36	1929	874	7.32	4.93	193	146	237	166	1947	245	0.619	35.4	188	108
200	120	6.3	29.3	37.4	1976	898	7.27	4.9	198	150	244	172	2040	255	0.613	34.1	187.4	107.4
200	120	8	36.5	46.4	2386	1079	7.17	4.82	239	180	298	209	2507	308	0.606	27.4	184	104
200	120	10	44.4	56.6	2806	1262	7.04	4.72	281	210	356	250	3007	364	0.597	22.5	180	100
200	120	12	50.9	64.9	3031	1368	6.84	4.59	303	228	395	278	3419	406	0.578	19.6	176	96
200	120	12.5	52.6	67	3099	1397	6.8	4.57	310	233	406	285	3514	416	0.576	19	175	95
250	150	5	30.1	38.4	3304	1508	9.28	6.27	264	201	320	225	3285	337	0.783	33.2	240	140
250	150	6	35.8	45.6	3886	1768	9.23	6.23	311	236	378	266	3886	396	0.779	27.9	238	138
250	150	6.3	37.2	47.4	4001	1825	9.18	6.2	320	243	391	276	4078	412	0.773	26.8	237.4	137.4
250	150	8	46.5	59.2	4886	2219	9.08	6.12	391	296	482	340	5050	504	0.766	21.5	234	134
250	150	10	57	72.6	5825	2634	8.96	6.02	466	351	582	409	6121	602	0.757	17.6	230	130
250	150	12	66	84.1	6458	2925	8.77	5.9	517	390	658	463	7088	684	0.738	15.2	226	126
250	150	12.5	68.3	87	6633	3002	8.73	5.87	531	400	678	477	7315	704	0.736	14.6	225	125
250	150	16	83.8	106.8	7660	3453	8.47	5.69	613	460	805	566	8713	823	0.718	11.9	218	118
260	180	5	33.2	42.4	4121	2350	9.86	7.45	317	261	377	294	4695	426	0.863	30.1	250	170
260	180	6.3	41.2	52.5	5013	2856	9.77	7.38	386	317	463	361	5844	523	0.853	24.3	247.4	167.4
260	180	8	51.5	65.6	6145	3493	9.68	7.29	473	388	573	446	7267	642	0.846	19.4	244	164
260	180	10	63.2	80.6	7363	4174	9.56	7.2	566	464	694	540	8850	772	0.837	15.8	240	160
260	180	12	73.5	93.7	8245	4679	9.38	7.07	634	520	790	615	10330	884	0.818	13.6	236	156

260	180	12.5	76.2	97	8482	4812	9.35	7.04	652	535	8158	635	10680	911	0.818	13.1	235	155
260	180	16	93.9	120	9923	5614	9.11	6.85	763	624	977	759	12890	1079	0.798	10.7	228	148
300	100	6	35.8	45.6	4477	842	10.2	4.3	318	168	411	188	2403	306	0.779	27.9	288	88
300	100	6.3	37.2	47.4	4907	868	10.2	4.28	327	174	425	194	2515	318	0.773	26.8	287.4	87.4
300	100	8	46.5	59.2	5978	1045	10	4.2	399	209	523	238	3080	385	0.766	21.5	284	84
300	100	10	57	72.6	7106	1224	9.9	4.11	474	245	631	285	3681	455	0.757	17.6	280	80
300	100	12	66	84.1	7808	1343	9.64	4	521	269	710	321	4177	508	0.738	15.2	276	76
300	100	12.5	68.3	87	8010	1374	9.59	3.97	534	275	732	330	4292	521	0.736	14.6	275	75
300	100	16	83.8	107	9157	1543	9.26	3.8	610	309	865	386	4939	592	0.718	11.9	268	68
300	150	6	40.5	51.6	6074	2080	10.8	6.35	405	277	500	309	4988	479	0.879	24.7	288	138
300	150	6.3	42.2	53.7	6266	2150	10.8	6.32	418	287	517	321	5234	499	0.873	23.7	287.4	137.4
300	150	8	52.8	67.2	7684	2623	10.7	6.25	512	350	640	396	6491	612	0.866	18.9	284	134
300	150	10	64.8	82.6	9209	3125	10.6	6.15	614	417	776	479	7879	733	0.857	15.4	280	130
300	150	12	75.4	96.1	10300	3498	10.4	6.03	687	466	883	546	9153	837	0.838	13.3	276	126
300	150	12.5	78.1	99.5	10590	3595	10.3	6.01	706	479	912	563	9452	862	0.836	12.8	275	125
300	150	16	96.4	123	12390	4174	10	5.83	826	557	1092	673	11330	1015	0.818	10.4	268	118
300	200	6	45.2	57.6	7370	3962	11.3	8.29	491	396	588	446	8115	651	0.979	22.1	288	188
300	200	6.3	47.1	60	7624	4104	11.3	8.27	508	410	610	463	8524	680	0.973	21.2	287.4	187.4
300	200	8	59.1	75.2	9389	5042	11.2	8.19	626	504	757	574	10630	838	0.966	16.9	284	184
300	200	10	72.7	92.6	11310	6058	11.1	8.09	754	606	921	698	12990	1012	0.957	13.8	280	180
300	200	12	84.8	108	12790	6854	10.9	7.96	853	685	1056	801	15240	1167	0.938	11.8	276	176
300	200	12.5	88	112	13180	7060	10.8	7.94	879	706	1091	828	15770	1204	0.936	11.4	275	175
300	200	16	109	139	15620	8340	10.6	7.75	1041	834	1319	1000	19220	1442	0.918	9.2	268	168
350	250	6	54.7	69.6	12460	7458	13.4	10.3	712	597	843	671	14550	967	1.18	18.3	338	238
350	250	6.3	57	72.6	12920	7744	13.3	10.3	738	620	876	698	15290	1010	1.17	17.5	337.4	237.4
350	250	8	71.6	91.2	16000	9573	13.2	10.2	914	766	1092	869	19140	1253	1.17	14	334	234
350	250	10	88.4	113	19410	11590	13.1	10.1	1109	927	1335	1062	23500	1522	1.16	11.3	330	230
350	250	12	104	132	22200	13260	13	10	1268	1061	1544	1229	27750	1770	1.14	9.65	326	226
350	250	12.5	108	137	22920	13690	12.9	9.99	1310	1095	1598	1272	28770	1830	1.14	9.3	325	225
350	250	16	134	171	27580	16430	12.7	9.81	1576	1315	1954	1554	35500	220	1.12	7.46	318	218
400	200	8	71.6	91.2	18970	6517	14.4	8.42	949	652	1173	728	15820	1133	1.17	14	384	184
400	200	12.5	108	137	27100	9260	14.1	8.22	1355	926	1714	1062	23600	1644	1.14	9.3	375	175
400	200	16	134	171	32550	11060	13.8	8.05	1627	1106	2093	1294	28930	1984	1.12	7.5	368	168
400	300	8	84.2	107	25120	16210	15.3	12.3	1256	1081	1487	1224	31180	1747	1.37	11.9	384	284
400	300	10	104	133	30610	19730	15.2	12.2	1530	1315	1824	1501	38410	2132	1.36	9.6	380	280
400	300	12	123	156	35280	22750	15	12.1	1764	1516	2122	1747	45530	2492	1.34	8.16	376	276
400	300	12.5	127	162	36490	23520	15	12	1824	1568	2198	1810	47240	2580	1.34	7.9	375	275
400	300	16	159	203	44350	28540	14.8	11.9	2218	1902	2708	2228	58730	3159	1.32	6.28	368	268